

Gauge-Fixed Physical First-Alias Observation Map

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Abstract

The folded noisy operator lives on $[0, 1]$, whereas its deterministic physical flat trace sums signed periodic points on $[-1, 1]$. Localizing the latter by $x \in J \subset [0, 1]$ therefore fails to partition the physical trace. We repair the observation map by localizing with $|x| \in J$. For every $n \geq 1$, absolute value is a multiplier-preserving bijection between the fixed points of the signed map f^n and those of the folded map $T^n = |f|^n$. For every measurable J and $n \geq 2$, the corresponding localized signed and folded noisy operator traces are exactly equal. At $n = 2$ the old convention misses an explicit negative periodic point and the exact weight $(4u_c - 3)^{-1}$. Freezing the earlier boundary, sibling, and far partition with explicit endpoint ownership then restores the exact physical raw-trace partition. In the Hardy full-trace constituent, the corrected first-alias coefficient satisfies $q_{\text{FT}} = \mathcal{B} + \mathcal{S} + \mathcal{R} + \mathcal{P} - \mathcal{A}$. It is not the modulus complement: the latter differs by the noisy-head/counterloop defect. Finally, moving a positive-measure fixed-point-free set between unfrozen windows produces a nonzero zero-sum slot change. This window-partition dependence is distinct from the exchange-observation gauge inside a fixed model shell. All results are exact finite observation identities or scoped negative statements; no slot asymptotic, trace replacement, determinant closure, or Riemann-hypothesis conclusion is obtained.

1 The data-type defect and the corrected object

Let $u_c \in (1, 2)$ be the root of

$$u^3 - 2u^2 + 2u - 2 = 0, \quad (1)$$

and put

$$I = [-1, 1], \quad I_+ = [0, 1], \quad f(x) = 1 - u_c x^2, \quad T(y) = |f(y)|. \quad (2)$$

Write

$$r = u_c - 1, \quad \lambda = 2u_c r, \quad b = u_c^{-1/2}. \quad (3)$$

The signed critical orbit and its folded version are

$$0 \mapsto 1 \mapsto -r \mapsto r \mapsto r, \quad 0 \mapsto 1 \mapsto r \mapsto r. \quad (4)$$

The point b is the unique cusp of T and $b \mapsto 0 \mapsto 1 \mapsto r$ under T , so it is not periodic. The signed map, its complete physical flat trace, and the exact deterministic/noisy folding are the objects fixed in RH-10, RH-14, and RH-15 [2, 7, 8].

The localized definition in RH-327 [9] instead combined a folded noisy operator on I_+ with

$$P_n^{\text{old}}(J) = \sum_{\substack{f^n(x)=x \\ x \in J}} \frac{1}{|1 - (f^n)'(x)|}, \quad J \subset I_+. \quad (5)$$

Since the sets $J \subset I_+$ cannot contain negative signed periodic points, their union does not in general recover the physical flat trace. The corrected deterministic observation is

$$P_n^{\text{abs}}(J) = \sum_{\substack{f^n(x)=x \\ |x| \in J}} \frac{1}{|1 - (f^n)'(x)|}. \quad (6)$$

The superscript denotes localization by absolute coordinate; the denominator already retains the usual absolute fixed-point weight.

2 Exact localized noisy folding

For $\sigma > 0$, let

$$\phi_\sigma(t) = \frac{1}{\sqrt{2\pi}\sigma} e^{-t^2/(2\sigma^2)} \quad (7)$$

and define the signed conditioned kernel on I by

$$Z_\sigma(x) = \int_{-1}^1 \phi_\sigma(z - f(x)) dz, \quad (8)$$

$$q_\sigma(x, z) = \frac{\phi_\sigma(z - f(x))}{Z_\sigma(x)}. \quad (9)$$

The folded kernel on I_+ is

$$p_\sigma(y, t) = q_\sigma(y, t) + q_\sigma(y, -t). \quad (10)$$

These are exactly the signed and folded row-normalized Gaussian kernels used in the source chain [3, 6, 7]. Let $\mathcal{K}_\sigma^s$ and $\mathcal{K}_\sigma^f$ act on backward observables in $L^2(I)$ and $L^2(I_+)$, respectively. Thus the first kernel variable is the source.

Both kernels are bounded and continuous on compact rectangles, so both operators are Hilbert–Schmidt. Their powers of order at least two are trace class [1]. If $J \subset I_+$ is measurable, write M_J^f for multiplication by $\mathbf{1}_J$ and M_J^s for multiplication by $\mathbf{1}_{\{|x| \in J\}}$.

Theorem 2.1 (Localized signed–folded noisy trace identity). *For every measurable $J \subset I_+$, every $\sigma > 0$, and every integer $n \geq 2$,*

$$\boxed{\operatorname{Tr}_{L^2(I)}(M_J^s(\mathcal{K}_\sigma^s)^n) = \operatorname{Tr}_{L^2(I_+)}(M_J^f(\mathcal{K}_\sigma^f)^n)}. \quad (11)$$

In particular, the multiplication operator marks the same cyclic source or basepoint in the two data types.

Proof. Let $E : L^2(I_+) \rightarrow L^2(I)$ be $(Eh)(x) = h(|x|)$ and let $A : L^2(I) \rightarrow L^2(I_+)$ be

$$(Ag)(y) = \int_I q_\sigma(y, z)g(z) dz.$$

Evenness of f and Z_σ gives

$$\mathcal{K}_\sigma^s = EA, \quad \mathcal{K}_\sigma^f = AE, \quad M_J^s E = E M_J^f. \quad (12)$$

The operator A is Hilbert–Schmidt and E is bounded. For $n \geq 2$, $E M_J^f(\mathcal{K}_\sigma^f)^{n-1}$ and A are Hilbert–Schmidt. Cyclicity for products of two Hilbert–Schmidt operators therefore yields

$$\begin{aligned} \operatorname{Tr}_I(M_J^s(\mathcal{K}_\sigma^s)^n) &= \operatorname{Tr}_I(E M_J^f(\mathcal{K}_\sigma^f)^{n-1} A) \\ &= \operatorname{Tr}_{I_+}(\mathcal{K}_\sigma^f M_J^f(\mathcal{K}_\sigma^f)^{n-1}) \\ &= \operatorname{Tr}_{I_+}(M_J^f(\mathcal{K}_\sigma^f)^n). \end{aligned}$$

□

There is also a direct sign-path reading. In the signed closed-loop integral, write $x_j = \varepsilon_j y_j$, where $y_j \geq 0$. Since the source sign drops out of q_σ , summing independently over every destination sign gives

$$\sum_{\varepsilon_{j+1}=\pm 1} q_\sigma(y_j, \varepsilon_{j+1} y_{j+1}) = p_\sigma(y_j, y_{j+1}).$$

The product over a cyclic sign path therefore folds lobe by lobe with no factor two left over. This loop argument is compatible with eq. (12); the operator factorization supplies the required trace typing even though M_J may be discontinuous.

Remark 2.2 (Why $n \geq 2$ is retained). Lidskii's theorem may express either unlocalized trace as a spectral sum once the relevant power is trace class. It does not prove the localized identity after M_J is inserted. The repository guarantees Hilbert–Schmidt operators and trace-class powers for $n \geq 2$, which is sufficient above. We make no $n = 1$ operator-trace claim. A separately defined diagonal integral at $n = 1$ would be a different statement.

3 The multiplier-preserving fixed-point bijection

Theorem 3.1 (Unique signed lift of every folded periodic basepoint). *For every integer $n \geq 1$, absolute value is a bijection*

$$|\cdot| : \text{Fix}(f^n) \longrightarrow \text{Fix}(T^n). \quad (13)$$

Every folded periodic point avoids the cusp b , and if $y = |x|$ under (13), then

$$(T^n)'(y) = (f^n)'(x). \quad (14)$$

Consequently fixed-point simplicity, marked-point multiplicity, minimal period, and the physical flat-trace weight are preserved.

Proof. If $f^n(x) = x$, the semiconjugacy $|f(x)| = T(|x|)$ gives $T^n(|x|) = |x|$. Injectivity follows from evenness: if $x, x' \in \text{Fix}(f^n)$ and $|x| = |x'|$, then $f(x) = f(x')$ and applying f^{n-1} gives $x = x'$.

Conversely, let $y_0 \in \text{Fix}(T^n)$ and set $y_j = T^j y_0$, with indices modulo n . Define

$$x_j = f(y_{j-1}). \quad (15)$$

Then $|x_j| = y_j$, and evenness gives $f(x_j) = f(y_j) = x_{j+1}$. Thus $x_0 \in \text{Fix}(f^n)$ and $|x_0| = y_0$, proving surjectivity and uniqueness.

No periodic orbit can meet b : after b the folded orbit is $0, 1, r, r, \dots$. It also cannot meet 0 or 1 . Hence all derivatives below exist. Since $x_j = \text{sgn}(x_j) y_j$ and f' is odd,

$$f'(x_j) = \text{sgn}(x_j) f'(y_j), \quad T'(y_j) = \text{sgn}(x_{j+1}) f'(y_j).$$

The two sign products agree after cyclic index shift, proving (14). The signed periodic points are repelling and simple by the exact Markov result of RH-10; the same now holds for their folded images. If folding reduced a minimal period, the unique lift of the shorter folded cycle would reduce the signed period as well, which is impossible. $\square$

Corollary 3.2 (Correct localized deterministic trace). *For every measurable $J \subset I_+$ and every $n \geq 1$,*

$$P_n^{\text{abs}}(J) = \sum_{\substack{T^n(y)=y \\ y \in J}} \frac{1}{|1 - (T^n)'(y)|}. \quad (16)$$

In particular, $P_n^{\text{abs}}(I_+) = P_n$, the complete signed physical flat trace.

3.1 An explicit order-two counterexample to the old binning

For $p(u) = u^3 - 2u^2 + 2u - 2$, one has $p'(u) = 3u^2 - 4u + 2 > 0$, $p(3/2) = -1/8$, and $p(25/16) = 233/4096$. Hence

$$\frac{3}{2} < u_c < \frac{25}{16}. \quad (17)$$

Polynomial division gives

$$f^2(x) - x = (f(x) - x)\{u_c^2 x^2 - u_c x + 1 - u_c\}. \quad (18)$$

The two in-domain roots of the second factor are

$$x_{\pm} = \frac{1 \pm \sqrt{4u_c - 3}}{2u_c}, \quad x_- < 0 < x_+, \quad (19)$$

and $f(x_-) = x_+$, $f(x_+) = x_-$. The first factor in (18) has roots r and $-1/(u_c r)$. By (17), $u_c r < 225/256 < 1$, so the latter root is below -1 . Thus r, x_-, x_+ are exactly the three physical points in $\text{Fix}(f^2)$. The two-cycle multiplier and its weight are

$$(f^2)'(x_{\pm}) = 4u_c^2 x_- x_+ = -4r, \quad w_c = \frac{1}{1 + 4r} = \frac{1}{4u_c - 3}. \quad (20)$$

At r , $(f^2)'(r) = \lambda^2$ and

$$w_r = \frac{1}{\lambda^2 - 1}. \quad (21)$$

Thus a partition of I_+ using (5) sums only $w_r + w_c$, while the physical trace is

$$P_2 = w_r + 2w_c. \quad (22)$$

The exact missing amount is w_c .

Lemma 3.3 (Exact rational separation for the frozen fixture). *The three points in $\text{Fix}(T^2)$ satisfy*

$$0 < \max\{|x_-|, r\} < b - \frac{1}{8} < b < x_+ < b + \frac{1}{8} < 1. \quad (23)$$

Moreover, none belongs to $[b - 1/10, b - 2/25]$.

Proof. The bracket (17) gives $4/5 < b < \sqrt{2/3} < 7/8$ and $r < 9/16$; in particular the two fixture windows lie inside I_+ . Also $|x_-| < 1/2$, because $\sqrt{4u_c - 3} < 1 + u_c$. Thus both low points are below $b - 1/8 > 27/40$.

Since $u_c > 1$, $\sqrt{4u_c - 3} > 2\sqrt{u_c} - 1$, and therefore $x_+ > b$. The bracket above gives $\sqrt{u_c} > 61/50$ and $\sqrt{4u_c - 3} < 181/100$. Consequently

$$0 < x_+ - b = \frac{1 + \sqrt{4u_c - 3} - 2\sqrt{u_c}}{2u_c} < \frac{37}{300} < \frac{1}{8}.$$

Finally, the left endpoint of the remaining test interval obeys $b - 1/10 > 7/10$, whereas $|x_-| < 1/2$ and $r < 9/16$; its right endpoint is below $b < x_+$. This proves the last assertion. $\square$

At 110-decimal working precision the reproduction values are

quantity	ordinary decimal reproduction
r	0.5436890126920763615708559718
x_+	0.9010184338065749680123723533
x_-	-0.2532195625455325825218740568
w_r	0.5501733230542994206834636063
w_c	0.3149848315929645801889119110

These decimals reproduce exact algebraic formulas; they are not interval certificates.

4 The frozen physical basepoint observation map

The RH-327 coordinates use the fold cusp b , a source center $s_{\sigma,k}$, and

$$q_b = \frac{b - s_{\sigma,k}}{\sqrt{\sigma}}, \quad q(x) = \frac{x - s_{\sigma,k}}{\sqrt{\sigma}}.$$

For fixed $\sigma > 0$, integer k , and $A > 0$, their boundary-owned windows simplify exactly to

$$J_{\sigma,k,A}^- = I_+ \cap [b - A\sqrt{\sigma}, b), \quad (24)$$

$$J_{\sigma,k,A}^+ = I_+ \cap [b, b + A\sqrt{\sigma}], \quad (25)$$

$$F_{\sigma,k,A} = I_+ \setminus (J_{\sigma,k,A}^- \cup J_{\sigma,k,A}^+). \quad (26)$$

The source center cancels from the endpoints, but the physical derivation and the labels remain those of RH-327. We require (σ, k, A) and every endpoint convention to be frozen before evaluating a trace or inspecting a periodic point [9].

For measurable $J \subset I_+$ and $n \geq 2$, define

$$L_{\sigma,n}(J) = \text{Tr}\left(M_J^f(\mathcal{K}_\sigma^f)^n\right). \quad (27)$$

By theorem 2.1, this is also the signed cyclic observation with $|x_0| \in J$. Fix $r_H = 0.85$ and put

$$\mathcal{B}_{\sigma,k,n} = r_H^{-n} \{L_{\sigma,n}(J^-) - P_n^{\text{abs}}(J^-)\}, \quad (28)$$

$$\mathcal{S}_{\sigma,k,n} = r_H^{-n} \{L_{\sigma,n}(J^+) - P_n^{\text{abs}}(J^+)\}, \quad (29)$$

$$\mathcal{R}_{\sigma,k,n} = r_H^{-n} \{L_{\sigma,n}(F) - P_n^{\text{abs}}(F)\}. \quad (30)$$

Theorem 4.1 (Gauge-fixed physical raw-trace partition). *For every frozen partition (24)–(26) and every $n \geq 2$,*

$$\boxed{r_H^{-n} \{\text{Tr}(\mathcal{K}_\sigma^f)^n - P_n\} = \mathcal{B}_{\sigma,k,n} + \mathcal{S}_{\sigma,k,n} + \mathcal{R}_{\sigma,k,n}.} \quad (31)$$

Proof. The three indicators sum to one, so trace linearity partitions the noisy trace. By corollary 3.2, the same disjoint sets partition every physical fixed point exactly once and their deterministic sums add to P_n . Subtract and apply the common Hardy scale. $\square$

For the archived reproduction choice $k = 2$, $\sigma = 1/4$, and $A = 1/4$,

$$J^- = [b - 1/8, b), \quad J^+ = [b, b + 1/8]. \quad (32)$$

By lemma 3.3, the folded order-two points $|x_-|, r, x_+$ lie respectively in F, F, J^+ . Hence

$$(P_2^{\text{abs}}(J^-), P_2^{\text{abs}}(J^+), P_2^{\text{abs}}(F)) = (0, w_c, w_r + w_c), \quad (33)$$

whereas the old positive- x convention gives $(0, w_c, w_r)$.

5 Strict partition dependence and the separate model gauge

The explicit endpoint convention makes the frozen slots well defined. It does not make them invariant under changing the observation windows.

Proposition 5.1 (Positive nontrivial window shift). *Fix $\sigma > 0$ and $n \geq 2$. Let J_1, J_2, J_3 be a measurable partition of I_+ and let $E \subset J_1$ have positive Lebesgue measure and contain no point of $\text{Fix}(T^n)$. Reassign E from J_1 to J_2 . If*

$$D_j = r_H^{-n} \{L_{\sigma,n}(J_j) - P_n^{\text{abs}}(J_j)\},$$

then the slot vector changes by

$$(D_1, D_2, D_3) \mapsto (D_1 - \delta_E, D_2 + \delta_E, D_3), \quad \delta_E = r_H^{-n} L_{\sigma,n}(E) > 0. \quad (34)$$

Its total is unchanged.

Proof. The deterministic term on E is zero. The folded Gaussian kernel is strictly positive, so the diagonal kernel of its n th power is strictly positive and

$$L_{\sigma,n}(E) = \int_E p_{\sigma}^{(n)}(y, y) dy > 0.$$

Additivity gives (34). $\square$

For (32), the set

$$E = [b - 1/10, b - 2/25) \subset J^- \quad (35)$$

has length $1/50$ and, by lemma 3.3, contains no folded period-two point. Moving it from J^- to F changes $(\mathcal{B}, \mathcal{S}, \mathcal{R})$ by $(-\delta_E, 0, +\delta_E)$.

This is *window-partition dependence*: it changes which basepoints the observable marks. It is not the exchange-observation gauge of RH-330 [5], which keeps one frozen model shell $\hat{\mathcal{S}} = \hat{\mathcal{X}} + \hat{\mathcal{O}}$ fixed while sending $\hat{\mathcal{X}} \mapsto \hat{\mathcal{X}} + t$ and $\hat{\mathcal{O}} \mapsto \hat{\mathcal{O}} - t$. The former is eliminated by freezing J^-, J^+, F before evaluation. The latter requires a physical identification of the internal shell split; without it only the aggregate $\hat{\mathcal{S}}$ is intrinsic.

6 The correctly typed first-alias coefficient

We now connect the repaired observation map to the exact RH-326 algebra [4]. Let $\lambda_-(\sigma)$ be the global noisy parity eigenvalue and define

$$c_{\sigma,n}^H = r_H^{-n} \{ \text{Tr}(\mathcal{K}_{\sigma}^f)^n - 1 - \lambda_-(\sigma)^n \}, \quad (36)$$

$$c_n^H = r_H^{-n} \{ P_n - 1 - (-1)^n \}. \quad (37)$$

For $k \geq 2$, let M_k be the archived boundary-cycle multiplier and put

$$\beta_k = \frac{|M_k|^{-1/(2k)}}{r_H}, \quad \beta = \frac{1}{r_H \sqrt{\lambda}}. \quad (38)$$

Retain the finite counterloop $s_{k,n}$, limiting pole moment, and deterministic numerator anchor

$$s_{k,n} = \beta_k^n \{ 2k \mathbf{1}_{2k|n} - 1 - (-1)^n \}, \quad (39)$$

$$p_n^{\text{pole}} = -2 \mathbf{1}_{2|n} \beta^n, \quad a_n^{\text{num}} = c_n^H - p_n^{\text{pole}}. \quad (40)$$

Define the global parity and alias packets

$$\mathcal{P}_{\sigma,n} = r_H^{-n} \{ (-1)^n - \lambda_-(\sigma)^n \}, \quad \mathcal{A}_{k,n} = s_{k,n} - p_n^{\text{pole}}. \quad (41)$$

Neither global scalar is localized: no Perron/parity projector and no time-Floquet sector is inserted into (28)–(30). The RH-326 full-trace residual is

$$e_{\sigma,k,n} := c_{\sigma,n}^H - s_{k,n} - a_n^{\text{num}}. \quad (42)$$

Theorem 6.1 (Gauge-fixed first-alias observation ledger). *At $n = 2k$, define*

$$q_{\text{FT},\sigma,k,n} := e_{\sigma,k,n} = c_{\sigma,n}^H - s_{k,n} - a_n^{\text{num}}, \quad (43)$$

$$\text{coefficient_type} = \text{hardy_full_trace_constituent}. \quad (44)$$

Then

$$\boxed{q_{\text{FT},\sigma,k,2k} = e_{\sigma,k,2k} = \mathcal{B}_{\sigma,k,2k} + \mathcal{S}_{\sigma,k,2k} + \mathcal{R}_{\sigma,k,2k} + \mathcal{P}_{\sigma,2k} - \mathcal{A}_{k,2k}.} \quad (45)$$

Proof. Equations (36)–(41) give the exact identity

$$c_{\sigma,n}^H - c_n^H = r_H^{-n} \{ \text{Tr}(\mathcal{K}_{\sigma}^f)^n - P_n \} + \mathcal{P}_{\sigma,n}.$$

Using $a_n^{\text{num}} = c_n^H - p_n^{\text{pole}}$ in (43) gives

$$q_{\text{FT}} = (c_{\sigma,n}^H - c_n^H) - (s_{k,n} - p_n^{\text{pole}}).$$

Apply theorem 4.1 and (41). $\square$

6.1 It is not the modulus complement

Let $h_{\sigma,n}$ be a noisy-head moment in the same Hardy normalization, put

$$\tau_{\sigma,n} = c_{\sigma,n}^H - h_{\sigma,n}, \quad d_{\sigma,k,n} = h_{\sigma,n} - s_{k,n}. \quad (46)$$

Then exact rearrangement gives

$$\boxed{\tau_{\sigma,n} - a_n^{\text{num}} = q_{\text{FT},\sigma,k,n} - d_{\sigma,k,n}.} \quad (47)$$

Thus $q_{\text{FT}} = e$ is the Hardy full-trace constituent. Exact equality with the modulus-complement coefficient requires $d = 0$. Substitution at a target asymptotic scale instead requires a separate theorem making d negligible at that scale. This is the fixed-order head-transport distinction retained in the earlier annular route; RH-334 supplies no such head theorem.

6.2 Exact rational fixture

The artifact freezes $k = 2$, $n = 4$, $r_H = 17/20$ and the exact data

$$(L^-, L^+, L^F) = (3/2, 3/4, 1/4), \quad (P^-, P^+, P^F) = (11/10, 2/5, 1/5),$$

with $\lambda_- = -3/4$, $s = 3/2$, and $p^{\text{pole}} = -2$. Exact rational arithmetic gives

quantity	exact value	quantity	exact value
$c_{\sigma,4}^H$	189375/83521	c_4^H	-48000/83521
a_4^{num}	119042/83521	$\mathcal{A}$	7/2
$\mathcal{B}$	64000/83521	$\mathcal{S}$	56000/83521
$\mathcal{R}$	8000/83521	$\mathcal{P}$	109375/83521
q_{FT}	-109897/167042	five-slot error	0

Taking $d = 5/12$ yields $\tau - a^{\text{num}} = -1076987/1002252 = q_{\text{FT}} - d$, again with exact zero error. This fixture checks algebra and types; it is not physical trace data.

7 Reproduction protocol

The executable artifact has four independent layers.

1. It solves (1) at 110 decimal working digits and evaluates the exact order-two roots, orbits, multipliers, and weights. The old positive- x mutation is rejected because its slots fail to add to P_2 .
2. A rational four-state signed matrix with identical rows on each sign pair is folded to a two-state matrix. Its two localized trace-square values agree exactly: 43/100 and 229/400.
3. On 32 Gauss–Legendre nodes in I_+ and the corresponding 64 symmetric signed nodes, it builds the signed and folded Gaussian Nyström matrices at $\sigma = 1/4$. Destination quadrature weights are included in the matrices. The localized order-two rows are

slot	signed trace	folded trace	absolute error
J^-	0.2339453602109879	0.2339453602109879	0
J^+	0.2619881093922252	0.2619881093922252	0
F	1.042468475597926	1.042468475597926	0

The zero is a finite symmetric-matrix distributivity identity, not a continuum certificate.

4. A separate positive quadrature over the set (35) returns $L_{\sigma,2}(E) = 0.033293751080257\dots$ and $\delta_E = 0.046081316374058\dots > 0$. Exact fractions verify both paths to (45) and (47). A wrong `coefficient_type` mutation is rejected.

All decimal rows are deterministic reproduction checks. They neither prove the continuum operator theorem—which was proved analytically above—nor provide interval enclosures or moving-order evidence.

8 Claim boundary and next admissible step

RH-334 proves an exact physical observation map and an exact correction to the RH-327 localized deterministic slots. It also proves two scoped negative statements: positive- x binning fails already at order two, and individual localized slots are not invariant under unfrozen window changes.

It does *not* estimate the sign or magnitude of $\mathcal{B}, \mathcal{S}, \mathcal{R}$, prove that $\mathcal{R} = o(H_k)$, or establish a moving-order localization theorem. It does not identify any RH-322–RH-324 or RH-333 forward probability law with the cyclic trace, identify a physical exchange parameter, replace an actual operator by a model, close the all-leg Duhamel sum, control the off-alias background or noisy head, or glue a determinant. The raw full-line affine obstruction of RH-333 remains a probability-path result, not a trace theorem.

A continuation must keep the windows and coefficient type frozen, estimate the three observable slots on one common first-alias clock, and prove every actual/model and head/off-alias bridge before invoking a full-trace criterion. No result here constructs a Hilbert–Polya operator, identifies a Riemann zero, proves a von Mangoldt trace formula or completed-zeta divisor equality, or proves the Riemann hypothesis. Gates A–E remain false/open.

References

- [1] Barry Simon. *Trace Ideals and Their Applications*. American Mathematical Society, 2 edition, 2005.
- [2] Bin Wang. Parity-extracted bulk scattering, 2026. Repository paper RH-15.
- [3] Bin Wang. Complement excursion self-energy, 2026. Repository paper RH-19.
- [4] Bin Wang. Parity-renormalized first-alias packet identity, 2026. Repository paper RH-326.
- [5] Bin Wang. Joint-cancellation full-trace transfer criterion, 2026. Repository paper RH-330.
- [6] Bin Wang. Branch-isolated gaussian return, 2026. Repository paper RH-18.
- [7] Bin Wang. Parity-renormalized long-cycle determinant, 2026. Repository paper RH-10.
- [8] Bin Wang. Square-root parity boundary layer, 2026. Repository paper RH-14.
- [9] Bin Wang. Neighboring-shell coupling and cancellation budgets, 2026. Repository paper RH-327.